Alicia in Wonderland

The Red & White Queens

Four-dimensional space cannot be entered, but it can be seen, by its shadows and by its slices. In 1900 Alicia Boole Stott, daughter of George Boole and a contemporary of Lewis Carroll, published the slices of the 600-cell, which she had learned to see without formal training. This book completes that picture: the White Queen 600-cell, with all 62 slice patterns in her four main directions, her shadows and her fifteen shells of cells; and her polar partner the Red Queen 120-cell, built exactly from her.

On the way, three small pieces close the gaps between icosahedra, and the shadows of four dimensions give new solids with equal triangular faces. At the end, the five pyramids that Aristotle said fill space turn out to be exactly the cores of the four-dimensional regular solids, seen in perspective.

DICTO is an artist in Osaka, Japan, with a background in architectural sculpture, stonecarving and Buddhist statuary. He builds geometric sculpture from recycled PET bottles with his own modular systems, B1NR and DICTOspheres. He is also the author of HexOct: The Laws of Stacking and The Golden Cycle.

◆ DICTO ◆



Alicia in Wonderland

The Red & White Queens

◆ DICTO ◆

Open access edition: DICTO DOI 10.5281/zenodo.23753305
Licence CC BY-SA 4.0 · dictospheres.com